

OMKAR KORADE

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Education

Indiana University Bloomington

Master of Science, Data Science; CGPA: 3.73/4

Bloomington, IN

Aug. 2024 – May 2026

Vivekanand Education Society's Institute of Technology

Bachelor of Technology, AI & Data Science; CGPA: 8.11/10

Mumbai, India

Aug. 2020 – May 2024

Technical Skills

Languages/Frameworks: Python (PyTorch, TensorFlow, Flask), R, SQL (MySQL, MongoDB), MCP, Google ADK

Data Engineering: ETL/ELT pipelines, Data Warehousing, Data Quality, Data Architecture, BigQuery, Snowflake

Technologies: Claude Code, Cursor, Openclaw, AWS (S3, EC2), GCP, MLFlow, AirFlow, BigQuery, Git

Visualization Tools: Power BI, Tableau, Matplotlib, Seaborn, Plotly, ggplot2, Pandas, NumPy, NLTK, Scikit-learn

Experience

Research Assistant

Indiana University Bloomington

Jan. 2026 - Present

Bloomington, Indiana

- Engineered an end-to-end facial analysis pipeline processing 1,200+ images with automated quality filtering and deep learning-based feature extraction, achieving a 98% processing success rate while maintaining full IRB compliance.
- Trained and validated a regression model on facial embeddings ($R^2=0.759$) that predicted biological aging markers beyond chronological age, with statistically significant results across three established aging benchmarks.
- Ran a systematic facial region analysis to identify which areas of the face contribute most to biological age prediction, surfacing insights that varied meaningfully across different aging measures.

Data Analyst

University Graduate School

May. 2025 – Present

Bloomington, Indiana

- Built retrieval-based AI chatbot for IU graduate schools, integrating a hybrid MCP server agent to supplement knowledge source responses with structured program links and contact info, cutting student information lookup time by 60%.
- Created keyword-based school routing logic with regex word-boundary matching to eliminate false substring matches on acronyms, improving query precision across ambiguous multi-school prompts.
- Hardened a PHP/MySQL admin portal managing PhD defense listings by eliminating SQL injection vulnerabilities, enforcing CAS authentication restricted to authorized IU Network IDs.

Machine Learning Intern

Trivia Softwares

Dec. 2023 – Feb. 2024

Mumbai, India

- Improved ML model prediction speed by 11% through advanced regression, classification, and clustering techniques, demonstrating problem-solving in large-scale dataset environments.
- Streamlined KPI monitoring by 33% via automated dashboards for 20+ metrics, enabling data-driven decision-making in real time.

Academic Projects

Alzheimer's Disease Classification via CNNs | Python, TensorFlow, OpenCV | GitHub

Nov. 2025 – Present

- Developed deep learning pipeline for classifying MRI brain scans into 4 Alzheimer's stages achieving 67.7% test accuracy on OASIS-1 dataset with 20,000 balanced images.
- Engineered robust data preprocessing pipeline resolving critical data leakage issues through patient-level splitting and similarity analysis; implemented strategic undersampling and augmentation to balance dataset from 83% majority class to equal 25% distribution.
- Architected efficient CNN with 3 convolutional blocks, batch normalization, dropout, and L2 regularization, reducing parameters by 40% while maintaining performance; integrated learning rate scheduling and early stopping for optimal convergence.

AI Meeting Companion | Python, Whisper API, NLTK | GitHub

Mar. 2025 – Jun. 2025

- Architected a real-time transcription system handling 6+ audio formats with 95.7% accuracy using Whisper API and Python audio processing (FFMPEG/pydub)
- Structured extracted data with Pydantic models and Instructor pattern, achieving 91.2% consistency in action item identification
- Developed hierarchical summarization combining TextRank algorithms and AI-powered abstractive methods, reducing information retrieval time by 40% and integrated fallback mechanisms ensuring 100% uptime during API outages